Fiscal Year:	FY 2021	**Task Last Updated:**	FY 02/25/2021
PI Name:	Dunbar, Bonnie Ph.D.		
Project Title:	Retired Astronauts Medical Data Repository (RADAR): a Feasibility Project		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2020 per TRISH (Ed., 5/28/2020) NOTE: End date changed to 6/30/2020 (original end date 5/31/2020) per E. Urquieta/TRISH (Ed., 10/3/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schmit, Cason Ph.D. (Texas A&M Research Foundation) Bloomfield, Susan Ph.D. (Texas A&M Research Foundation)		
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Task Description:	Focused Investigation Project Understanding long-term impacts of space exploration on human health requires tracking crewmember medical status well after retirement. Lifetime Surveillance of Astronaut Health (LSAH) collects data on United States and Canadian crewmembers, but does not collect data on other international crewmembers. To reduce risk, the sample size must be enlarged. Retired Astronauts Medical Data Repository (RADAR) will investigate the feasibility for developing an international retired astronaut data base. The long-term goal of this work is to create a retired astronaut medical data repository housed at Texas A&M University-College Station to capture data that will otherwise be lost and to greatly increase the “n” for future research. Ultimately, data for all retired crew in this repository would be made available for use by qualified investigators to study the long-term health impact of exposure to the space environment in order to mitigate health risks for future crew. All data from US and Canadian retirees would be made available to NASA’s LSAH. For this one-year feasibility project, we propose to undertake critical activities required to assure the eventual success of such a data repository. Hence, our short-term goals include (1) Project Definition: conducting thorough collection of stakeholder input on issues related to collecting medical data; (2) Minimal Viable Prototype (MVP): adapting an existing software platform to make medical data collection user-friendly and easy; creating a minimally viable prototype for data collection and repository in which to bank those data; (3) Consenting Issue Resolution: developing a “willing cohort” of retired crew members by obtaining prospective consent to said data collection; (4) Pilot Data Collection: running a pilot data collection trial; and (5) Strategic Plan: generating a strategic plan complete with a realistic supporting budget for the full development of a retired astronaut medical data repository. 1. A “willing” cohort of retired astronauts (both domestic and international) that has been socialized to sharing of medical data, “real-world” data from wearable technologies/ genomics services, and end-of-life reports. 2. A well-defined prospective consenting procedure that will be useful for retired crew across multiple countries. 3. A comprehensive strategic plan for moving to full implementation with this Retired Astronaut Medical Data Repository, including: a. Roadmap for entire project ; b. Evaluation of lessons learned from feasibility project, including pilot testing of “best fit” platform that is HIPAA-compliant, with defined interoperability requirements ; c. High level cost analysis with technology/personnel requirements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Retired Astronauts Medical Data Repository (RADAR) has demonstrated the feasibility of collecting retired international astronaut data in order to increase the population size for research into the effects of space flight on long duration health. In fact, the processes described within the report could be expanded in the future to the commercial astronauts, not under the purview of NASA or any other agency. Our specific deliverables and their impacts follow: 1. We have verified, contrary to some expectations, that there is a willing cohort of retired astronauts (both domestic and international) that has been socialized to sharing of medical data, real-world data from wearable technologies/genomics services, and end-of-life reports. 2. We have verified that most of this population would be willing to share their post retirement medical data as long as privacy can be assured and if they have a "user friendly" interface technology, such as a Personal Health Record (PHR) App. 3. We wrote and published a Product Requirements Document (PRD) for the selection of a PHR App to be used by retired international (and domestic) astronauts. 4. Using the PRD, we selected a Commercial Minimum Viable Product (MVP) PHR app and tested it with a United States (US) Cohort: ----In order to use this selected App, modifications would be required by the vendor, but they are willing to work with NASA/Translational Research Institute for Space Health (TRISH) to this end. ----In order for US astronauts to use this App, one of the menu destinations would need to be NASA/Flight Medicine AND NASA/Flight Medicine may need to update their portal. 5. We evaluated the legal environments for the European Union, Canada, Russia, Japan, and India and determined that there should be no obstacles to retired international astronauts for sharing their own data with the destination of their choice. 6. We have negotiated with the Association of Space Explorers (ASE) for the development of an organization ID card which would allow individuals to designate themselves available for autopsy with data to be shared with "to be determined" (TBD) (in the case of US, this would be NASA/LSAH). This is analogous to an Organ Donor Card. However, this effort will probably require further legal discussions. 7. We have developed a top-level strategic plan which would take us to the next steps for the PHR technology development and for the conceptual design of an international data repository.
	Understanding long-term impacts of space exploration on human health requires tracking crewmember medical status well after retirement. Lifetime Surveillance of Astronaut Health (LSAH) collects data on US and Canadian crewmembers, but does not collect data on other international crewmembers. To help to reduce the future risk to human space flight, the sample size must be enlarged in order to ensure more statistically reliable conclusions. The urgency for this effort is particularly acute since the numbers of American astronauts flying per year is declining while the average age (over 60) of retired US astronauts is increasing. Retired Astronauts Medical Data Repository (RADAR) was designed to investigate the feasibility for developing an international retired astronaut data base in order to increase the sample size. The possible opportunity for collecting this data was offered by the international retired astronaut cohort themselves through a policy statement they generated from the Association of Space Explorers (ASE) in 2018 (https://). ASE includes all international astronauts,

Task Progress:	numbering more than 400 from 32 nations. Defined Aims, Project Findings, and Future Work: 1. PROJECT DEFINITION: "Collect stakeholder input on proposed data repository and willingness to use electronic personal health record platforms for sharing medical data. This includes: (a) Surveys/focus groups at Association of Space Explorers Congress in October, 2019, in Houston; and (b) Surveys at the International Space Medicine Summit (ISMS) 2019 in Houston. It is essential to understand what will incentivize retired crew to regularly use an electronic personal health record platform -- what is the value of such a product to them personally (organizing their own medical data) and how it can easily transmit personal medical data to an eventual data repository." FINDINGS: After collecting stakeholder data at all proposed venues, we found broad consensus on the need to collect and share this data between investigators and crew, but there are many remaining questions on the location and construct of the "repository," and the mechanisms for collecting and sharing the data. Some of these questions are addressed in our key findings but many more answers will require a follow-on investigation. 2. DEVELOPING A MINIMAL VIABLE PROTOTYPE (MVP) FOR COLLECTING DATA: "Adapt an existing commercial Personal Health Record (PHR) platform for use in collecting medical data from retired crew, including interfaces on the back end with a data repository prototype. (a) Develop a Product requirement definition (PRD); (b) Landscape existing platforms; and (c) Finalize platform architecture, including HIPAA compliant features, interoperability requirements, etc." FINDINGS: We developed a detailed Product Requirements Document (PRD), designed from the perspective of the retired astronaut, for product review, and then down-selected five commercially available Personal/Electronic Health Platforms for evaluation by a cohort of US astronauts. Both the "App" design and its usability were evaluated by survey. Implementation is promising. 3. CONSENTING ISSUE RESOLUTION: "Create a willing cohort by obtaining prospective consent from retired crew to collect medical, end-of-life, and real-world data; (a) Generate prospective consenting documents compliant with US/Canadian medical privacy law for use with MVP for pilot data collection; and (b) Explore international medical privacy law impacting on crew from Russia, the European Union, and Japan." FINDINGS: (a) The international astronaut community is largely willing to use a software platform/technology for sharing data and (b) we found no obvious legal barriers for international retired astronauts to share their own medical data. International Law reviews included European Union, Russia, Japan, and India. In addition, the ASE has offered to work with NASA and the international partners to develop an ID card for members with consent to autopsy upon death (similar to organ donations). 4. PILOT DATA COLLECTION: "Encourage stakeholder buy-in and test MVP platform by collecting real-world data from retired crew. The MVP platform will be tested with a limited number of stakeholders, likely limited to US/Canadian retired crew (to minimize issues on consenting)." FINDING: A MVP Platform (PHR) was evaluated by survey with a limited US cohort of retired astronauts. Although there was broad support for the concept of a mobile PHR which connected individuals to their health providers and allowed the individuals to store and move data from provider to provider, from state to state, and internationally, this PHR did have some limitations which are described in the report. However, the Company is very interested in working with TRISH/NASA in customizing its platform for retired astronauts and moving it into international domains. This would not only help international astronauts share data but also optimize the transmittal of US data to LSAH/NASA. It was also clear, in this time of COVID, that portable personal health record (PHR) technology is in demand, that there are few technology providers in the market place, and that customizing a platform for the collection of international astronaut data could be a positive and visible pathfinder. 5. PREPARE STRATEGIC PLAN FOR NEXT PHASE: "The year's activities will culminate in the development of a comprehensive strategic plan to bring back to NASA/TRISH in a full proposal." FINDING/CONCLUSION: A key RADAR Project Deliverable is a top-level strategic plan for the next phase of investigation into the feasibility and early execution of this effort. This will form the basis for a follow on proposal, which will capture many of the findings reported in the full report, and include collaborative technology development for both the PHR platform as well as the "repository" for this data.
Bibliography Type:	Description: (Last Updated: 07/13/2023)
Articles in Peer-reviewed Journals	Bloomfield SA, Dunbar BJ, Schmit CD, Sawyer AJ, Charles JB. "Developing an international database on long-term health effects of spaceflight." Acta Astronautica. 2022 Sep 1;198:347-53. https://doi.org/10.1016/j.actaastro.2022.06.006 , Sep-2022